

ATPE - Adaptive Torque & Power Engine

Concept Whitepaper v1.0

Project Phoenix

Document status: Conceptual / simulation-validated - no hardware prototype measured

Publication date: July 2026

Classification: External distribution permitted with simulation disclaimer

Disclaimer: All performance figures in this document are engineering projections from physics-based computer simulation unless explicitly marked measured. Prefix external claims with ?simulation shows...? Hardware validation is required before any commercial or regulatory decision.

1. What It Is

The Adaptive Torque & Power Engine (ATPE) - also called the Adaptive Modular Linear Combustion Engine (AMLCE) - is a crankless, camless, modular free-piston linear generator for series-hybrid vehicles. Combustion drives opposed pistons linearly; integrated linear alternators produce DC electricity on a 400-800 V bus. Traction motors drive the wheels.

The central engineering insight: match displacement to demand at the cylinder level, not the engine level. Instead of one fixed-displacement block with binary cylinder deactivation, ATPE uses three size classes (tiers) of independently controlled combustion cartridges. A supervisory controller activates the smallest tier set that covers the generation setpoint, keeping each firing module near its optimal brake-specific fuel consumption (BSFC) band.

ATPE does not turn the wheels. It is a highly efficient electricity generator that runs at quasi-steady load while faster buffers (inertial rotor and battery) handle transients.

2. Reference Architecture

2.1 Three-tier cartridge stack (Phase 1 reference vehicle)

Tier	Role	Units	Displacement / unit	Max electrical (kW, tier total)	Target thermal efficiency
1 - Micro	Cruise, idle, light city	4	100 cc	30	44%
2 - Medium	Highway, moderate load	2	300 cc	80	41%
3 - Large	Peaks, grades, towing	2	750 cc	120	38%

Total stack (Phase 1): 8 cartridges, ~230 kW peak electrical output.

Tier selection is discrete and additive: activate from the smallest tier upward until demand is met. When multiple tiers fire together, fuel and power are blended using ****energy-weighted tier fill**** - each active tier contributes in proportion to its electrical output and thermal efficiency.

Production vision (PHOENIX-X12): cartridges arranged in a ring (X4 through X16 layouts) with mixed tier counts per ring position.

2.2 Single cartridge module

Each cartridge is self-contained:

- Opposed pistons (50 mm total stroke, ± 25 mm per side)
- Electronic valves - no camshaft
- Integrated linear alternator -> DC bus
- Bounce chamber / gas spring for piston return
- Local pressure and position sensing

There is no crankshaft and no shared camshaft.

2.3 Variable-geometry controls (crankless advantage)

Mechanism	Purpose
Electronic valve timing	Torque at low speed vs power at high speed
Variable compression ratio	High CR for efficiency; low CR for knock-limited boost
Variable stroke	On-demand expansion ratio (Atkinson/Miller-like)
Variable intake geometry	Runner length tuning per operating point
Tier activation	Generalised cylinder deactivation by size class

3. Operating Modes

Mode	Generation setpoint	Active tiers	Notes
Off	0 kW	None	Battery and buffer cover demand (urban EV)
Micro-only	0-30 kW	Tier 1	Highest thermal efficiency
Micro + medium	30-110 kW	Tiers 1-2	Highway charge-sustaining sweet spot
Full stack	>110 kW	Tiers 1-3	Large tier active $\leq 15\%$ of drive time
Degraded	Reduced cap	Remaining units	One cartridge offline - graceful degradation

Charge-sustaining state-of-charge target: 55%

EV-only floor SoC: 25%

4. Control Philosophy

ATPE generation is the slow loop (10-100 ms) in the unified powertrain controller.

The generation setpoint is:

$P_{gen}^* = \text{low-pass-filtered positive demand} + \text{SoC correction}$

with a filter time constant of approximately 5 seconds so the engine does not chase transients. Fast mismatch is handled by the PCMRITMS inertial buffer first (1-5 ms), then the battery.

Decision priorities:

1. Can the battery alone cover demand at acceptable SoC? -> ATPE off.
2. Does Tier 1 alone suffice? -> Micro cartridges only.
3. Demand exceeds Tier 1? -> Add Tier 2, then Tier 3 as needed.
4. Acceleration spike detected? -> Buffer discharges; ATPE setpoint rises on the slow loop.
5. Thermal / wear management -> rotate firing pattern across units within a tier.

5. Efficiency - Simulation Results

5.1 Engine-level comparison vs conventional crankshaft ICE

Metric	Conventional 2.0 L turbo	ATPE (simulation)
Part-load efficiency	20-28%	38-44% (tier-weighted)
Peak thermal efficiency	38-42%	42-47% (micro tier)
Mechanical losses (crank, cam)	8-12%	~2-3%
Displacement matching	Fixed block	Per-tier additive
Idle fuel	Moderate	~0 (combustion stops)

5.2 Vehicle fuel economy - AWD SUV reference (simulation)

Identical vehicle mass, drag, and tyres; only the powertrain differs.

Drive cycle	ATPE (L/100 km)	Conventional ICE (L/100 km)	Saving
Urban (charged battery)	0.00	0.00	EV-dominant
Highway (charge-sustaining)	4.46	7.67	42%
Tow + grade	6.91	11.92	42%
Mixed	2.35	3.26	28%
WLTP-class reconstruction	4.61	7.30	37%

5.3 CO₂ - AWD SUV (simulation)

Cycle	ATPE (g/km)	Conventional ICE (g/km)
Highway	103	177
Mixed	54	75
WLTP-class	106	169

5.4 Fleet-wide mixed-cycle fuel (simulation)

Vehicle body	Fuel (L/100 km)	Running cost (EUR /km)	Lifecycle CO ₂ (g/km)
Hatchback	0.55	0.039	44
Sedan	0.64	0.040	47
Crossover	1.11	0.048	60
AWD SUV	2.61	0.072	102
Van / MPV	2.95	0.077	112
Pickup	3.86	0.092	137

Uncertainty bands (5th-95th percentile) on AWD SUV running cost: EUR 0.066-0.091/km.

6. Fault Tolerance (simulation)

Engineering requirement: the vehicle must remain driveable with ****one combustion cartridge offline****.

Result (simulation): all six vehicle bodies pass 9 of 9 performance checks (acceleration, top speed, 20% gradeability, efficiency) in 18 of 18 fault scenarios (one cylinder removed from each tier, across all bodies).

Fault scenario	Peak power remaining (kW)	ERS checks
Tier 1 micro cylinder offline	222	9/9 PASS
Tier 2 medium cylinder offline	190	9/9 PASS
Tier 3 large cylinder offline	170	9/9 PASS

7. Virtual Single-Cartridge Bench (simulation acceptance criteria)

Before hardware is built, a 48-cell virtual bench matrix defines lab acceptance criteria:

Parameter	Value
Grid	4 speeds x 4 loads x 3 tiers = 48 cells
Overall pass rate (simulation)	100% (48/48)
Speeds tested	1800, 2200, 2600, 3000 rpm
Loads tested	25%, 50%, 75%, 100%

Sweet-spot operating point (Tier 2 medium, 2600 rpm, 75% load, simulation):

Metric	Value	90% confidence interval
Stroke	50.0 mm	-
Peak electrical power	58.1 kW	52.8-62.8 kW
Electric efficiency	49.6%	47-51%
IMEP	4.2 bar	3.9-4.6 bar
Bench acceptance checks	6/6 pass	-

Bearing runout target (hardware): ≤ 0.03 mm

Peak power target per cartridge at scale (hardware): ≥ 78 kW

8. Multi-Cartridge Ring Scaling (simulation)

Ring layouts from X4 to X16 were evaluated with one-, two-, and three-tier mixes.

Key finding (simulation): no homogeneous single-tier layout passes all nine vehicle performance targets. A multi-tier stack is required.

Layout category	Example mix (micro/medium/large)	ERS result	Highway fuel (L/100 km)
Single-tier micro only	14/0/0	8/9 FAIL	4.31
Single-tier medium only	0/4/0	7/9 FAIL	4.62
Single-tier large only	0/0/3	7/9 FAIL	4.99
Two-tier	1/4/0	9/9 PASS	4.58
Three-tier (Phase 1 reference)	4/2/2	9/9 PASS	4.46
Three-tier (X12 storyboard)	10/1/1	9/9 PASS	4.31

Rated electrical output scales from 105 kW (single-tier micro) to 478 kW (X16 three-tier).

9. Engineering Challenges

Challenge	Simulation status	Hardware requirement
Combustion timing without crank TDC	Physics surrogate model	Optical/pressure sensing per cartridge
Piston resonance / return	RPM-linked cycle model	Opposed-piston or active EM return
Bearing runout	Target <=0.03 mm specified	Displacement probes on magnetic bearings
NVH from async firing	Not modelled in detail	Acoustic test cell
Ring packaging	Concept renders only	Production CAD
Thermal zoning	Simplified model	Rig thermocouples + FEA

10. Comparison to Prior Art

Prior art	What it has	What ATPE adds
Toyota VCM / GM AFM	Binary cylinder on/off	Multi-tier sizing + free-piston
Achates opposed-piston	Opposed geometry, no crank	Modular ring + linear generation
Aquarius / Libertine FPE	Single free-piston generator	Scalable multi-tier architecture
Infiniti VC-Turbo	Variable compression	VC + free stroke + no crankshaft

11. Development Roadmap

Phase	Timeline	Deliverable
Virtual bench (complete)	Done	48-cell matrix, uncertainty bands
Single cartridge rig	Months 0-12	Measured pressure, stroke, kW
Stable free-piston	Months 6-18	Runout <=0.03 mm; stability envelope
Linear generator	Months 12-18	>=78 kW peak per cartridge
Multi-cartridge ring	Months 18-30	X8-X12 synchronization

Mule vehicle	Months 30-42	Compare highway fuel to 4.46 L/100 km target
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Seed-phase funding (illustrative): \$8-15M over 18 months for two bench prototypes with measured efficiency data.

12. Summary

ATPE replaces the crankshaft engine with a modular ring of free-piston linear generators sized in three tiers. Simulation shows approximately 28% less fuel than a conventional 2.0 L turbo on the same SUV, zero idle burn, and **fault tolerance with one cartridge offline** - provided hardware achieves the assumed thermal and electrical efficiencies.

The virtual 48-cell bench matrix defines what the first laboratory rig must measure before any commercial claim is upgraded from ?simulation shows? to ?measured.?

Companion documents: PCMRITMS Whitepaper v1.0; Project Phoenix Integrated System Whitepaper v1.0.